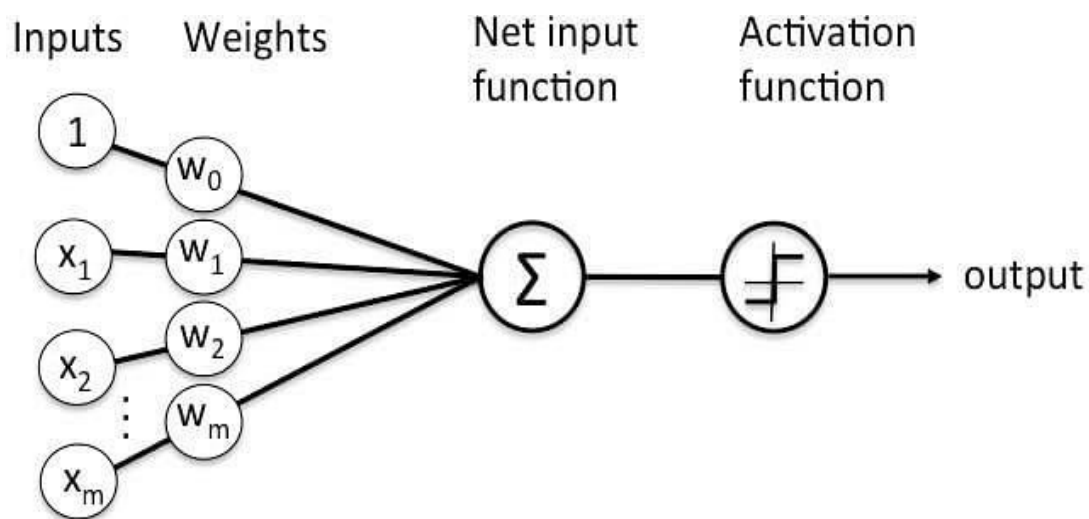


## Perceptron

Perceptron was introduced by Frank Rosenblatt in 1957. He proposed a Perceptron learning rule based on the original MCP neuron. A Perceptron is an algorithm for supervised learning of binary classifiers. This algorithm enables neurons to learn and processes elements in the training set one at a time.



### Types of Perceptron:

1. Single layer: Single layer perceptron can learn only linearly separable patterns.
2. Multilayer: Multilayer perceptrons can learn about two or more layers having a greater processing power.

The Perceptron algorithm learns the weights for the input signals in order to draw a linear decision boundary.

Note: Supervised Learning is a type of Machine Learning used to learn models from labeled training data. It enables output prediction for future or unseen data. Let us focus on the Perceptron Learning Rule in the next section.

## Perceptron in Machine Learning

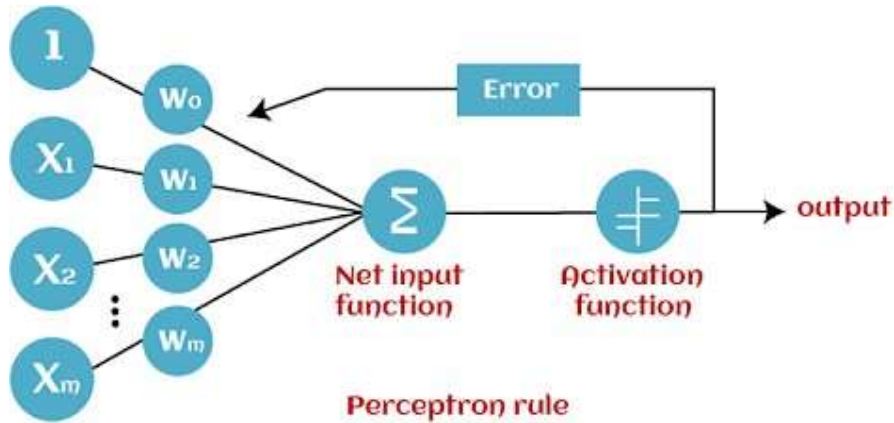
The most commonly used term in Artificial Intelligence and Machine Learning (AIML) is Perceptron. It is the beginning step of learning coding and Deep Learning technologies, which consists of input values, scores, thresholds, and weights implementing logic gates. Perceptron is the nurturing step of an Artificial Neural Link. In 19h century, Mr. Frank Rosenblatt invented the Perceptron to perform specific high-level calculations to detect input data capabilities or business intelligence. However, now it is used for various other purposes.

### What is the Perceptron Model in Machine Learning?

A machine-based algorithm used for supervised learning of various binary sorting tasks is called Perceptron. Furthermore, Perceptron also has an essential role as an Artificial Neuron or Neural link in detecting certain input data computations in business intelligence. A perceptron model is also classified as one of the best and most specific types of Artificial Neural networks. Being a supervised learning algorithm of binary classifiers, we can also consider it a single-layer neural network with four main parameters: input values, weights and Bias, net sum, and an activation function.

### How Does Perceptron Work?

AS discussed earlier, Perceptron is considered a single-layer neural link with four main parameters. The perceptron model begins with multiplying all input values and their weights, then adds these values to create the weighted sum. Further, this weighted sum is applied to the activation function 'f' to obtain the desired output. This activation function is also known as the step function and is represented by 'f.'



This step function or Activation function is vital in ensuring that output is mapped between (0,1) or (-1,1). Take note that the weight of input indicates a node's strength. Similarly, an input value gives the ability to shift the activation function curve up or down.

Step 1: Multiply all input values with corresponding weight values and then add to calculate the weighted sum. The following is the mathematical expression of it:

$$\sum w_i * x_i = x_1 * w_1 + x_2 * w_2 + x_3 * w_3 + \dots + x_n * w_n$$

Add a term called bias 'b' to this weighted sum to improve the model's performance.

Step 2: An activation function is applied with the above-mentioned weighted sum giving us an output either in binary form or a continuous value as follows:

$$Y = f(\sum w_i * x_i + b)$$

## Types of Perceptron models

We have already discussed the types of Perceptron models in the Introduction. Here, we shall give a more profound look at this:

1. **Single Layer Perceptron model:** One of the easiest ANN (Artificial Neural Networks) types consists of a feed-forward network and includes a threshold transfer inside the model. The main objective of the single-layer perceptron model is to analyze the linearly separable objects with binary outcomes. A Single-layer perceptron can learn only linearly separable patterns.
2. **Multi-Layered Perceptron model:** It is mainly similar to a single-layer perceptron model but has more hidden layers.

**Forward Stage:** From the input layer in the on stage, activation functions begin and terminate on the output layer.

**Backward Stage:** In the backward stage, weight and bias values are modified per the model's requirement. The backstage removed the error between the actual output and demands originating backward on the output layer. A multilayer perceptron model has a greater processing power and can process linear and non-linear patterns. Further, it also implements logic gates such as AND, OR, XOR, XNOR, and NOR.

### Advantages:

- A multi-layered perceptron model can solve complex non-linear problems.
- It works well with both small and large input data.
- Helps us to obtain quick predictions after the training.
- Helps us obtain the same accuracy ratio with big and small data.

Disadvantages:

- In multi-layered perceptron model, computations are time-consuming and complex.
- It is tough to predict how much the dependent variable affects each independent variable.
- The model functioning depends on the quality of training.

Characteristics of the Perceptron Model

1. It is a machine learning algorithm that uses supervised learning of binary classifiers.
2. In Perceptron, the weight coefficient is automatically learned.
3. Initially, weights are multiplied with input features, and then the decision is made whether the neuron is fired or not.
4. The activation function applies a step rule to check whether the function is more significant than zero.
5. The linear decision boundary is drawn, enabling the distinction between the two linearly separable classes +1 and -1.
6. If the added sum of all input values is more than the threshold value, it must have an output signal; otherwise, no output will be shown.

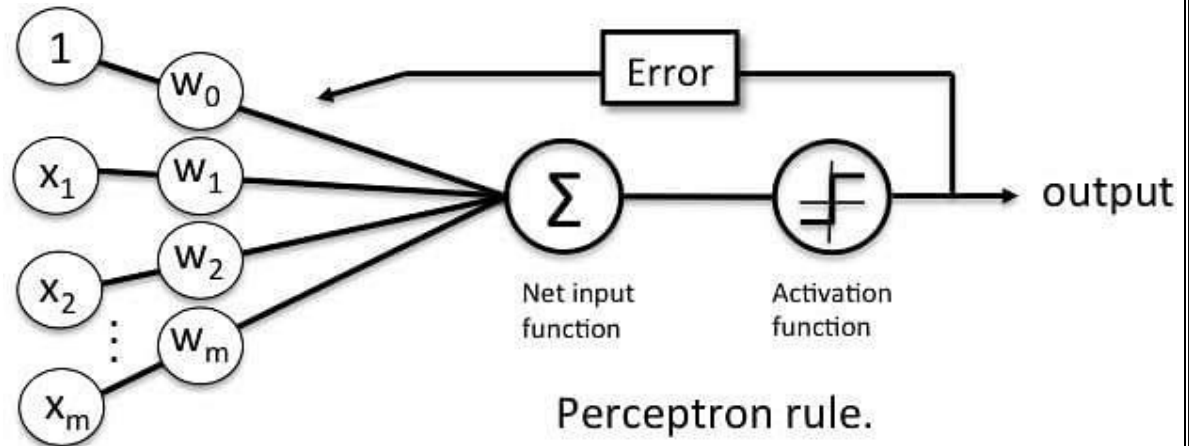
Limitation of Perceptron Model

The following are the limitation of a Perceptron model:

1. The output of a perceptron can only be a binary number (0 or 1) due to the hard-edge transfer function.
2. It can only be used to classify the linearly separable sets of input vectors. If the input vectors are non-linear, it is not easy to classify them correctly.

## Perceptron Learning Rule

Perceptron Learning Rule states that the algorithm would automatically learn the optimal weight coefficients. The input features are then multiplied with these weights to determine if a neuron fires or not.



The Perceptron receives multiple input signals, and if the sum of the input signals exceeds a certain threshold, it either outputs a signal or does not return an output. In the context of supervised learning and classification, this can then be used to predict the class of a sample.

Next up, let us focus on the perceptron function.

## Perceptron Function

Perceptron is a function that maps its input “x,” which is multiplied with the learned weight coefficient; an output value  $f(x)$  is generated.

$$f(x) = \begin{cases} 1 & \text{if } w \cdot x + b > 0 \\ 0 & \text{otherwise} \end{cases}$$

In the equation given above:

- “w” = vector of real-valued weights
- “b” = bias (an element that adjusts the boundary away from origin without any dependence on the input value)
- “x” = vector of input x values

$$\sum_{i=1}^m w_i x_i$$

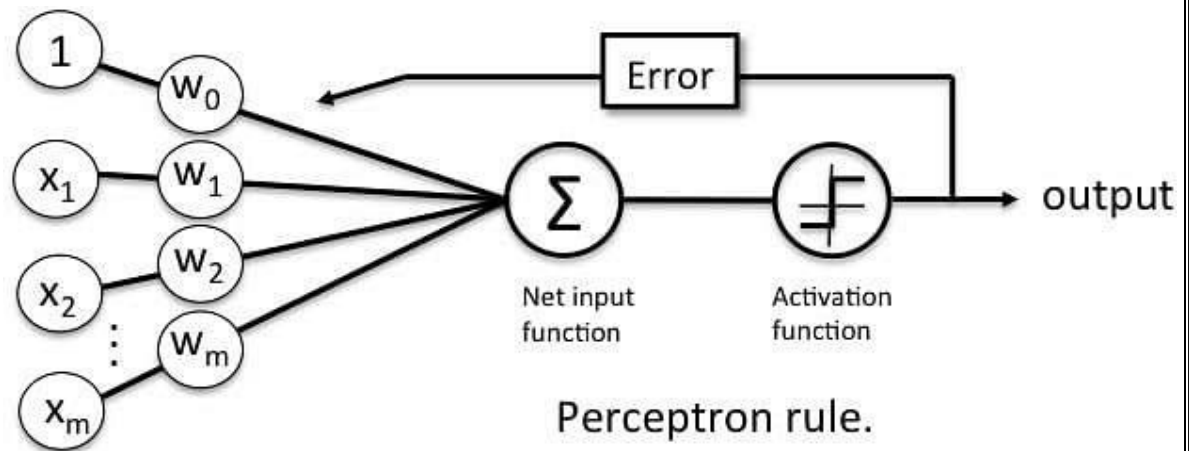
- “m” = number of inputs to the Perceptron

The output can be represented as “1” or “0.” It can also be represented as “1” or “-1” depending on which activation function is used.

Let us learn the inputs of a perceptron in the next section.

### Inputs of a Perceptron

A Perceptron accepts inputs, moderates them with certain weight values, then applies the transformation function to output the final result. The image below shows a Perceptron with a Boolean output.



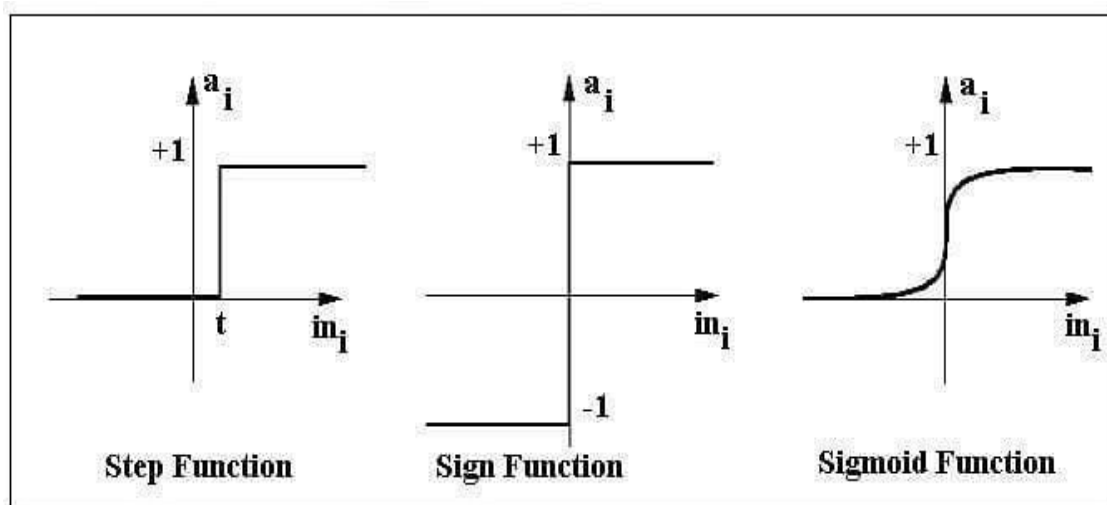
A Boolean output is based on inputs such as salaried, married, age, past credit profile, etc. It has only two values: Yes and No or True and False. The summation function " $\Sigma$ " multiplies all inputs of " $x$ " by weights " $w$ " and then adds them up as follows:

$$w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$$

In the next section, let us discuss the activation functions of perceptrons.

### Activation Functions of Perceptron

The activation function applies a step rule (convert the numerical output into +1 or -1) to check if the output of the weighting function is greater than zero or not.



For example:

If  $\sum w_i x_i > 0 \Rightarrow$  then final output “o” = 1 (issue bank loan)

Else, final output “o” = -1 (deny bank loan)

Step function gets triggered above a certain value of the neuron output; else it outputs zero. Sign Function outputs +1 or -1 depending on whether neuron output is greater than zero or not. Sigmoid is the S-curve and outputs a value between 0 and 1.

Output of Perceptron

Perceptron with a Boolean output:

Inputs:  $x_1 \dots x_n$

Output:  $o(x_1 \dots x_n)$

$$o(x_1, \dots, x_n) = \begin{cases} 1 & \text{if } w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n > 0 \\ -1 & \text{otherwise} \end{cases}$$

Weights:  $w_i \Rightarrow$  contribution of input  $x_i$  to the Perceptron output;

$w_0 \Rightarrow$  bias or threshold

If  $\sum w_i x_i > 0$ , output is +1, else -1. The neuron gets triggered only when weighted input reaches a certain threshold value.

$$o(\vec{x}) = \text{sgn}(\vec{w} \cdot \vec{x})$$

$$\text{sgn}(y) = \begin{cases} 1 & \text{if } y > 0 \\ -1 & \text{otherwise} \end{cases}$$

An output of +1 specifies that the neuron is triggered. An output of -1 specifies that the neuron did not get triggered.

“sgn” stands for sign function with output +1 or -1.

### Error in Perceptron

In the Perceptron Learning Rule, the predicted output is compared with the known output. If it does not match, the error is propagated backward to allow weight adjustment to happen.

Let us discuss the decision function of Perceptron in the next section.

### Perceptron: Decision Function

A decision function  $\phi(z)$  of Perceptron is defined to take a linear combination of  $x$  and  $w$  vectors.

$$\mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_m \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}$$

The value  $z$  in the decision function is given by:

$$Z = w_1 x_1 + \dots + w_m x_m$$

The decision function is +1 if  $z$  is greater than a threshold  $\theta$ , and it is -1 otherwise.

$$\phi(z) = \begin{cases} 1 & \text{if } z \geq \theta \\ -1 & \text{otherwise} \end{cases}$$

This is the Perceptron algorithm.

Bias Unit

For simplicity, the threshold  $\theta$  can be brought to the left and represented as  $w_0 x_0$ , where  $w_0 = -\theta$  and  $x_0 = 1$ .

$$Z = w_0 x_0 + w_1 x_1 + \dots + w_m x_m = \mathbf{w}^T \mathbf{x}$$

The value  $w_0$  is called the bias unit.

The decision function then becomes:

$$\phi(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

Output:

The figure shows how the decision function squashes  $w^T x$  to either +1 or -1 and how it can be used to discriminate between two linearly separable classes.

